Bilkent University Electrical and Electronics Department

EE102-01 Lab 3 Report:

Combinational Logic Circuit

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Purpose

The aim of lab is to learn how to design and implement a combinational circuit on breadboard using physical gates.

Design Specification

A function signal generator, logic gate components and a counter is used to realize the lab. The equipment to visualize the results is a probe, an oscilloscope, LEDs and resistors. For count mode, MR', CEP, CET, PE' pins of counter are connected to the DC voltage supply. The four outputs of counter are visualized by connecting them to the LEDs. Moreover, they are connected to AND logic gate whose output is connected to OR gate (Figure 1). The ultimate output is connected to both oscilloscope and a LED.

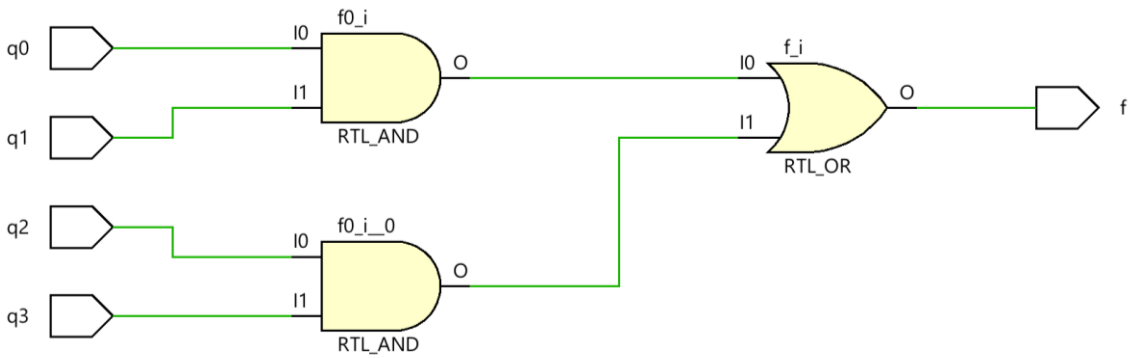

Figure 1: RTL schematic of the logic circuit

Methodology

To learn functionality and pins, datasheets are main tool to realize lab. To make appropriate connections, before starting, a plan is essential. The function signal generator is used with 1Hz square waves as a clock input of 4-bit counter (74 HC/HCT 163). As declared in function table of the component for count mode, we used "h" for MR', CEP, CET, PE'. Because we do not need D inputs, we connected them to the common ground. The Q outputs are connected to LEDs to visualize counting.
The gates whose inputs are the q outputs of 4-bit counter are a quad 2-input AND gate (74 LS/HC 08) and a quad 2-input OR gate (74 LS/HC 32). Obtained signals from counter goes to AND gates and ANDs output to OR. The output f is connected to a LED and the oscilloscope. Therefore, f = (Q0 * Q1) + (Q2 * Q3).

Results

The results from the waveform and output LED (Figure 2) are as expected from the logic function f = (Q0 * Q1) + (Q2 * Q3) which correlated with the truth table (Table 1) and observations through LEDs (Figure 2.0-16).

	Q3	Q2	Q1	Q0	Q0 * Q1	Q2 * Q3	f(Q3, Q2, Q1, Q0)
0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0
2	0	0	1	0	0	0	0
3	0	0	1	1	1	0	1
4	0	1	0	0	0	0	0
5	0	1	0	1	0	0	0
6	0	1	1	0	0	0	0
7	0	1	1	1	1	0	1
8	1	0	0	0	0	0	0
9	1	0	0	1	0	0	0
10	1	0	1	0	0	0	0
11	1	0	1	1	1	0	1
12	1	1	0	0	0	1	1
13	1	1	0	1	0	1	1
14	1	1	1	0	0	1	1
15	1	1	1	1	1	1	1

Table 1: Truth table of logic gates

Figure 2: The waveform of the logic function “f”

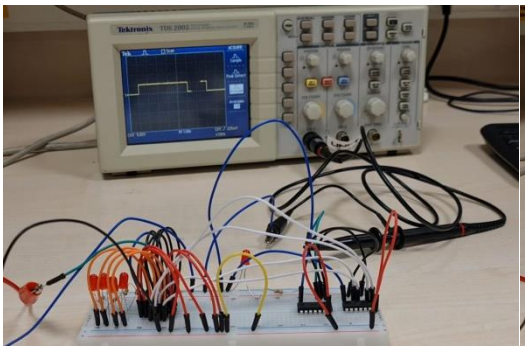
Figure 2.0: f(0,0,0,0) = 0

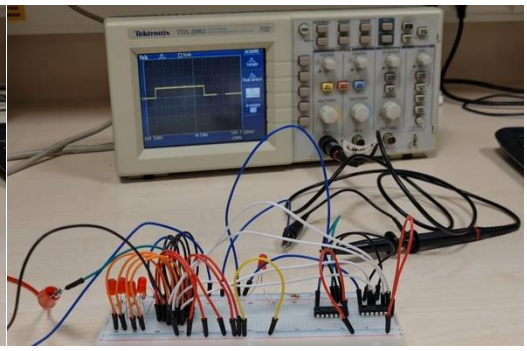
Figure 2.1: f(0,0,0,1) = 0

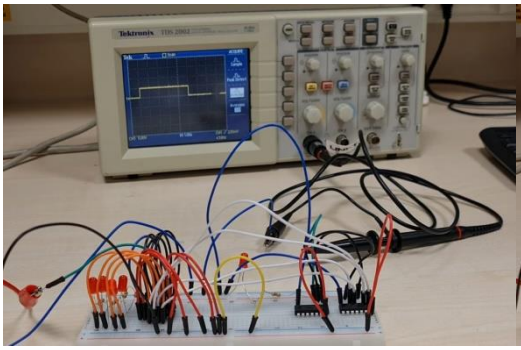
Figure 2.2: f(0,0,1,0) = 0

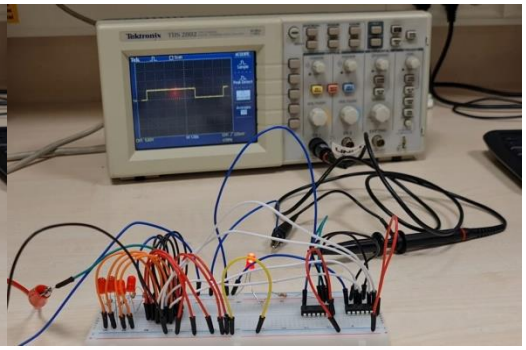
Figure 2.3: f(0,0,1,1) = 1

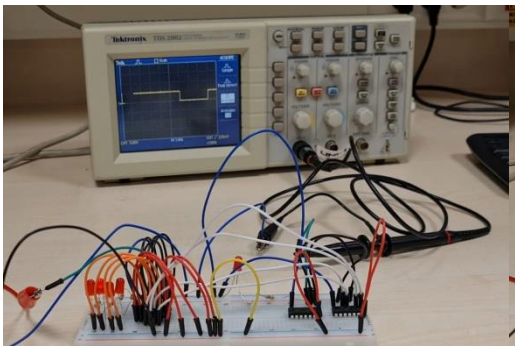
Figure 2.4: f(0,1,0,0) = 0

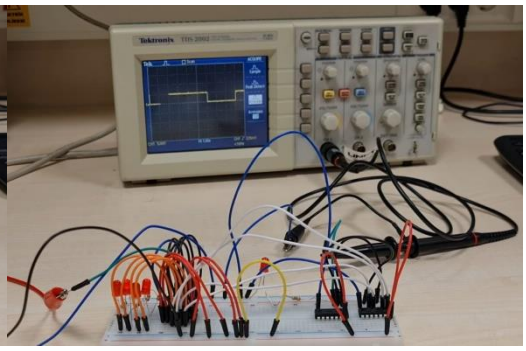
Figure 2.5: f(0,1,0,1) = 0

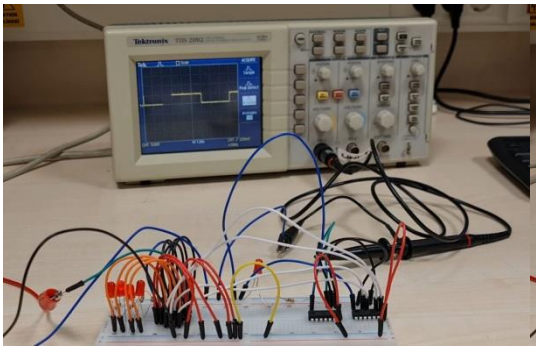
Figure 2.6: f(0,1,1,0) = 0

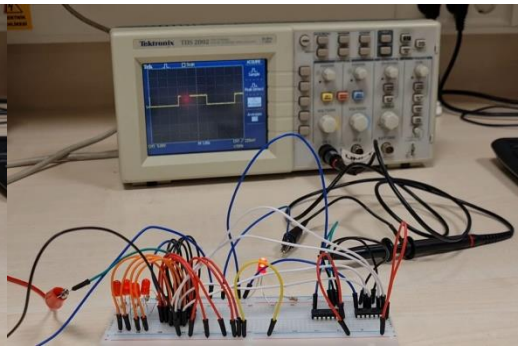
Figure 2.7: f(0,1,1,1) = 1

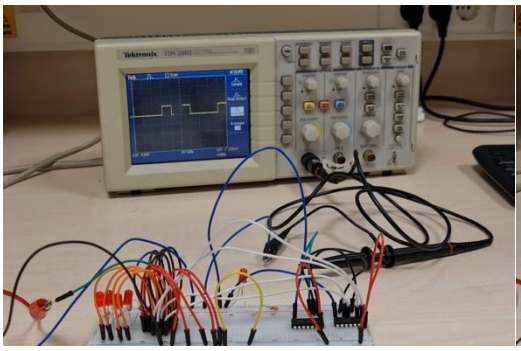
Figure 2.8: f(1,0,0,0) = 0

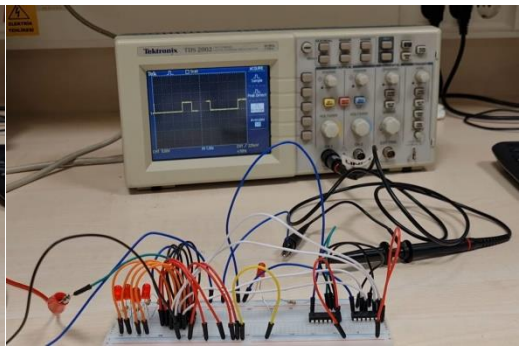
Figure 2.9: f(1,0,0,1) = 0

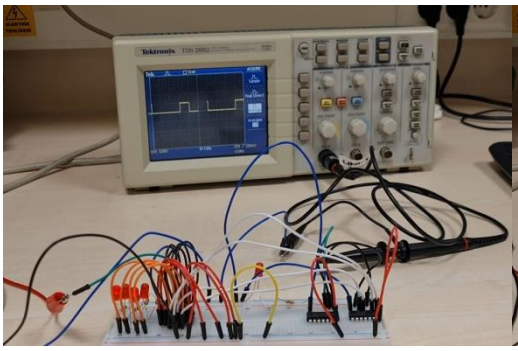
Figure 2.10: f(1,0,1,0) = 0

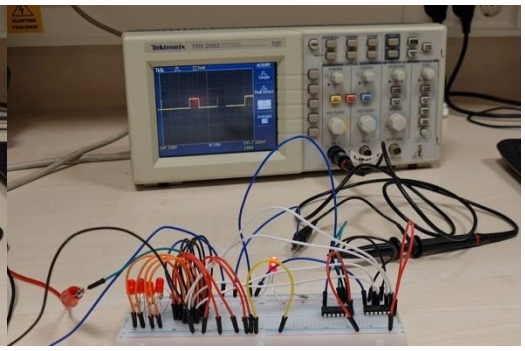
Figure 2.11: f(1,0,1,1) = 1

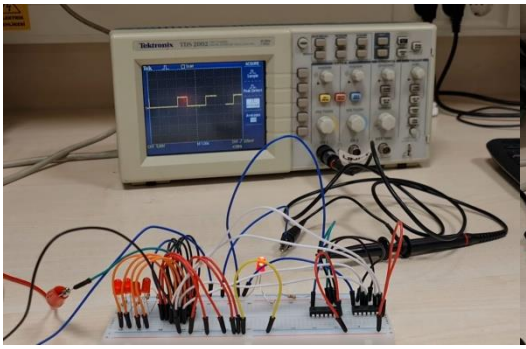
Figure 2.12: f(1,1,0,0) = 1

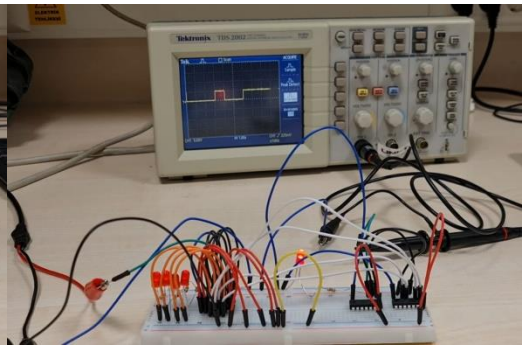
Figure 2.13: f(1,1,0,1) = 1

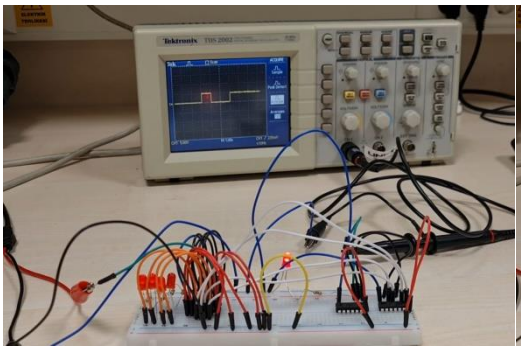
Figure 2.14: f(1,1,1,0) = 1

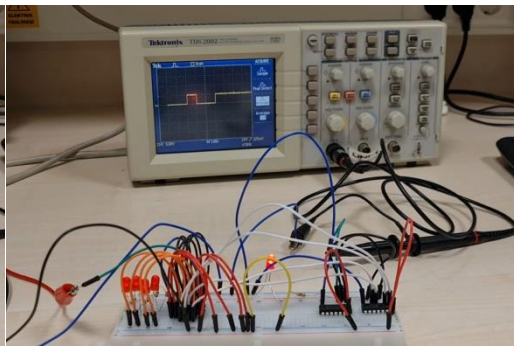
Figure 2.15: f(1,1,1,1) = 1

Conclusion

In this lab, a combinational circuit is designed on a breadboard. Using the 4-bit counter, we make observation through LEDs. Moreover, datasheets are examined in order to use components, which increases reading ability of datasheets. By doing so, inputs are adjusted to counting mode. The outputs of counter are inputs of logic circuit which has AND and OR gates. The logic circuit is connected to both the oscilloscope and a LED. By observing them, we compare the results with truth table which are as expected.

Furthermore, a 12-bit counter is made for entertainment (Figure 3). We demonstrate TC pin's functionality which becomes high (1) when the counter reaches 1111 (binary 15) and both CEP (Count Enable Parallel) and CET (Count Enable Trickle) are high. We connect TC pin of a counter to clock pin of another counter, and repeat to another. Therefore, we are able to count to 2^12-1 rather than 2^4-1.

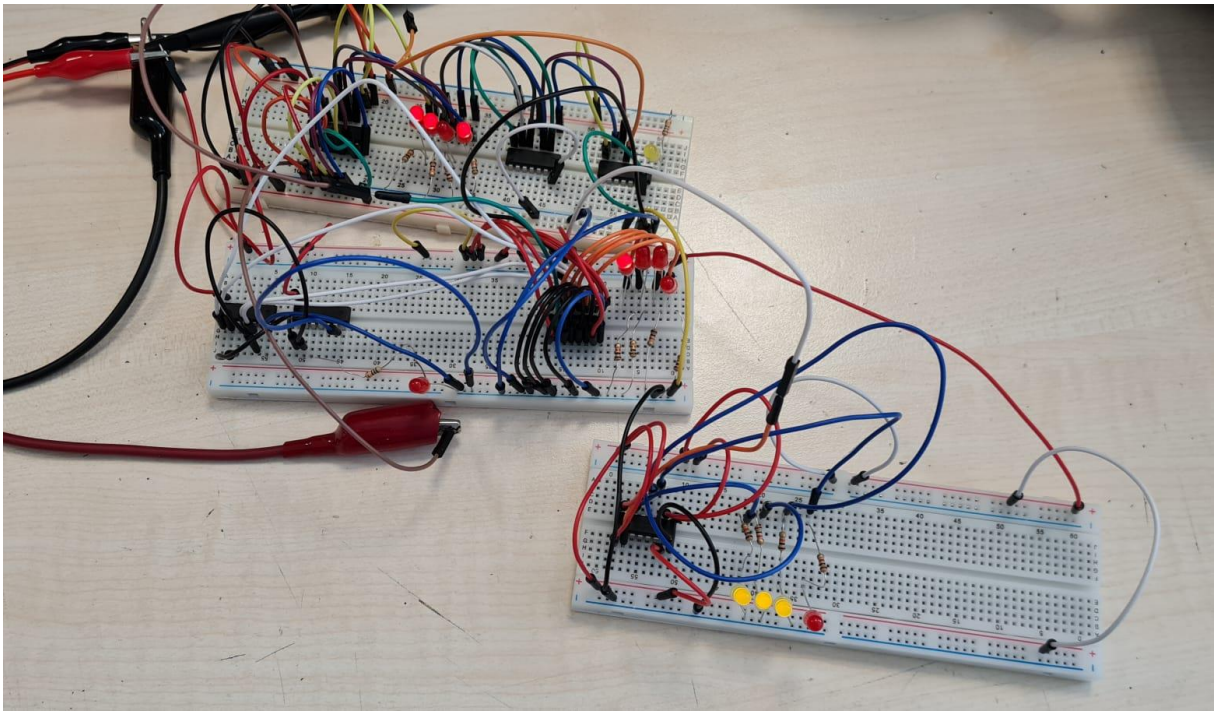

Figure 3: The 12-bit counter